Gaussian Mixture Models: Tutorial Handout (Additional)

AI 500 - Foundations of AI

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1 Introduction

Gaussian Mixture Models (GMMs) are probabilistic models that represent data as a combination of multiple Gaussian (normal) distributions. They are widely used in machine learning for tasks like clustering, density estimation, and image segmentation. Unlike simpler methods that assign each data point to exactly one cluster, GMMs provide a "soft" assignment by calculating the probability that each point belongs to each component of the mixture.

Why GMMs? Imagine analyzing pixel intensities in an image with a clear foreground and background. The histogram of pixel values might show two distinct peaks—one for darker background pixels and another for brighter foreground pixels. A single Gaussian distribution cannot capture this bimodal behavior, but a mixture of two Gaussians can model it naturally.

The challenge with GMMs is that we don't know in advance which data points belong to which Gaussian component. This is where the Expectation-Maximization (EM) algorithm comes in—an elegant iterative method for estimating both the component assignments and the parameters of each Gaussian.

2 Gaussian Distribution + Relevant Probability Concepts

2.1 The Univariate Gaussian Distribution

Before diving into mixtures, let's review the single Gaussian (normal) distribution. A Gaussian distribution is characterized by two parameters:

- $\boldsymbol{\mu}$: The mean, which determines the center/location of the distribution
- $\boldsymbol{\sigma}$: The standard deviation, which controls the spread/width of the distribution
- $\boldsymbol{\sigma^2}$: The variance (standard deviation squared)

The probability density function (pdf) of a Gaussian distribution is:

$$g(x|\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Breaking down the notation:

- $g(x|\mu, \sigma)$ means "the probability density of observing value x, given parameters μ and σ"
- The fraction $\frac{1}{\sigma\sqrt{2\pi}}$ is a normalization constant ensuring the total area under the curve equals 1
- The exponential term $e^{-\frac{(x-\mu)^2}{2\sigma^2}}$ creates the characteristic bell shape
- $(x-\mu)^2$ measures the squared distance from the mean—the further x is from μ, the lower the probability

Key properties:

- The distribution is symmetric around the mean μ
- About 68% of values fall within $\mu \pm \sigma$
- About 95% fall within $\mu \pm 2\sigma$
- The peak of the curve occurs at $x = \mu$

2.2 Probability Density vs. Probability

For continuous distributions like the Gaussian, $g(x|\mu, \sigma)$ is a **density**, not a probability. Probabilities are obtained by integrating the density over an interval:

$$P(a \leq x \leq b) = \int_a^b g(x|\mu, \sigma)dx$$

The density can exceed 1 (especially when σ is small), but the integral over all possible values always equals 1.

2.3 Maximum Likelihood Estimation (MLE)

Given a dataset, how do we find the best-fitting Gaussian parameters? We use Maximum Likelihood Estimation, which finds the parameters that make the observed data most probable.

For a single Gaussian with data points $x_1, x_2, ..., x_n$, the maximum likelihood estimates are:

$$\mu_{MLE} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

$$\sigma^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu)^2$$

These are simply the sample mean and sample variance—intuitive estimators that we'll extend to the mixture case.

3 Mixture Model Intuition

3.1 What is a Mixture Model?

A mixture model assumes that the data comes from multiple underlying distributions, but we don't know which distribution generated each data point. It's like having several overlapping populations mixed together.

Real-world analogy: Imagine measuring the heights of people in a room. If the room contains both adults and children, the overall height distribution would be bimodal (two peaks). We could model this as a mixture: a Gaussian for adult heights and another for child heights.

3.2 The Mixture of Gaussians Concept

A Gaussian Mixture Model with K components combines multiple Gaussian distributions:

- Each component i has its own parameters: mean μ_i and standard deviation σ_i
- Each component has a **weight** w_i representing the proportion of data from that component
- The weights satisfy: $\sum_{i=1}^{K} w_i = 1$ and $0 \leq w_i \leq 1$ for all i

Interpretation of weights: If $w_1 = 0.7$ and $w_2 = 0.3$ in a 2-component GMM, then roughly 70% of the data comes from the first Gaussian and 30% from the second.

3.3 Visual Intuition

Consider a histogram of image pixel intensities:

- **Unimodal histogram**: Can be modeled by a single Gaussian
- **Bimodal histogram**: Shows two peaks, suggesting two underlying populations (e.g., foreground and background pixels). This requires a mixture of two Gaussians.

The GMM flexibility allows it to approximate complex, multimodal distributions that a single Gaussian cannot capture.

4 Mathematical Formulation

4.1 The GMM Probability Density Function

For a 2-component Gaussian Mixture Model, the probability density function is:

$$g(x|w_1, \mu_1, \sigma_1, \mu_2, \sigma_2) = w_1 \cdot g(x|\mu_1, \sigma_1) + w_2 \cdot g(x|\mu_2, \sigma_2)$$

Where:

- w_1 and w_2 are the mixing weights with $w_1 + w_2 = 1$ (note: $w_2 = 1 - w_1$)
- $g(x|\mu_i, \sigma_i)$ is the Gaussian pdf for component i

Expanding this fully:

$$g(x) = w_1 \cdot \frac{1}{\sigma_1\sqrt{2\pi}} e^{-\frac{(x-\mu_1)^2}{2\sigma_1^2}} + w_2 \cdot \frac{1}{\sigma_2\sqrt{2\pi}} e^{-\frac{(x-\mu_2)^2}{2\sigma_2^2}}$$

4.2 Example

A specific 2-component GMM with equal weights:

$$g(x) = 0.5 \times \frac{1}{\sqrt{2\pi(20)^2}} e^{-\frac{(x-100)^2}{2(20)^2}} + 0.5 \times \frac{1}{\sqrt{2\pi(20)^2}} e^{-\frac{(x-180)^2}{2(20)^2}}$$

Parameters:

- $w_1 = w_2 = 0.5$ (equal mixing)
- $\mu_1 = 100$, $\sigma_1 = 20$ (first Gaussian centered at 100)
- $\mu_2 = 180$, $\sigma_2 = 20$ (second Gaussian centered at 180)

This creates two overlapping bell curves centered at intensities 100 and 180, each with the same spread.

4.3 The Parameter Estimation Problem

Given: A dataset $X = \{x_1, x_2, ..., x_n\}$ (e.g., pixel intensities)

Find: The parameters $\Theta = \{w_1, \mu_1, \sigma_1, \mu_2, \sigma_2\}$ that best fit the data

The Challenge: If we knew which data points belonged to which Gaussian, we could simply estimate parameters for each subset separately (using MLE). But we don't know the assignments! This is called a "latent variable problem"—the component labels are hidden.

Why is this hard? It's a circular dependency:

- To estimate parameters, we need to know which points belong to which component
- To assign points to components, we need to know the parameters

The EM algorithm breaks this circular dependency through iteration.

5 Expectation-Maximization (EM) Algorithm

The EM algorithm is a general-purpose iterative method for finding maximum likelihood estimates when there are latent (hidden) variables. For GMMs, the latent variables are the component assignments.

5.1 Algorithm Overview

The EM algorithm alternates between two steps:

1. **E-Step (Expectation):** Assume the parameters are known. Compute the probability that each data point belongs to each component.
2. **M-Step (Maximization):** Assume the assignments are known. Update the parameters to maximize the likelihood.

By iterating these steps, the algorithm converges to a local maximum of the likelihood function.

5.2 Initialization

Before starting, we must initialize the parameters. Common strategies include:

- Random initialization
- k-means clustering followed by computing means and variances
- Domain knowledge (e.g., for image segmentation, initialize one component to low intensities and another to high)

Important: EM is sensitive to initialization and may converge to local maxima. Running with multiple initializations and choosing the best result is common practice.

5.3 E-Step: Computing Responsibilities

Given current parameters: $w_1, \mu_1, \sigma_1, \mu_2, \sigma_2$

Goal: For each data point x, compute how likely it is to belong to each component.

5.3.1 Step 1: Compute the Likelihood

For each data point x and each component i, compute the **likelihood**:

$$L_i(x) = w_i \cdot g(x|\mu_i, \sigma_i)$$

Notation breakdown:

- $L_i(x)$ is the weighted probability density of point x under component i
- This combines the component weight w_i with the Gaussian density $g(x|\mu_i, \sigma_i)$

For component $i = 1, 2$:

$$L_i(x) = w_i \cdot \frac{1}{\sigma_i\sqrt{2\pi}} e^{-\frac{(x-\mu_i)^2}{2\sigma_i^2}}$$

5.3.2 Step 2: Compute the Total Likelihood

Sum the likelihoods across all components:

$$s_x = \sum_{j=1}^{2} L_j(x) = L_1(x) + L_2(x)$$

Interpretation: s_x is the total probability density of observing x under the mixture model. It's the denominator we'll use for normalization.

5.3.3 Step 3: Compute Responsibilities (Posterior Probabilities)

The **responsibility** $P_i(x)$ is the probability that point x was generated by component i:

$$P_i(x) = \frac{L_i(x)}{s_x} = \frac{L_i(x)}{\sum_{j=1}^{2} L_j(x)}$$

Notation clarification:

- $P_i(x)$ is also written as $P(i|x)$ in some texts, meaning "probability of component i given observation x"
- This is a **posterior probability** computed using Bayes' theorem
- $0 \leq P_i(x) \leq 1$ and $P_1(x) + P_2(x) = 1$

Intuition: If $P_1(x) = 0.9$ and $P_2(x) = 0.1$, then point x is much more likely to belong to component 1. This is a "soft" assignment—we don't force x into one component; instead, we acknowledge uncertainty.

5.3.4 E-Step for Image Data

When working with images, we process:

- x**:** A vector of all pixel intensities (size $|X|$ = number of pixels)
- **Output:** For each component i, a vector of responsibilities $P_i(x_j)$ for each pixel x_j

If we have a 100×100 image, $|X| = 10,000$ pixels, and we compute a $2 \times 10,000$ matrix of responsibilities.

5.3.5 E-Step for Histogram Data

For computational efficiency with images, we can work with histograms:

- Instead of individual pixels, use histogram bins (e.g., 256 bins for grayscale intensities 0-255)
- Let $\hat{H}(x)$ be the count of pixels with intensity x
- Compute responsibilities once per bin: $\hat{P}_i(x)$ for $x \in \{0, 1, ..., 255\}$
- Weight updates by histogram counts: $\hat{P}_i(x) = H(x) \cdot P_i(x)$

This reduces computation from millions of pixels to hundreds of bins.

5.4 M-Step: Updating Parameters

Given: The responsibilities $P_i(x)$ for all data points and components

Goal: Update the parameters w_i, μ_i, σ_i to maximize the likelihood

The M-step uses weighted versions of the standard mean and variance formulas, where each point is weighted by its responsibility to each component.

5.4.1 Update Weight

The new weight w_i' is the average responsibility for component i:

$$w_i' = \frac{1}{|X|} \sum_{x \in X} P_i(x)$$

Interpretation: This is the proportion of data "assigned" to component i. If most points have high responsibility for component 1, then w_1' will be large.

For histograms:

$$w_i' = \frac{1}{|X|} \sum_{x \in X} \hat{P}_i(x)$$

where $\hat{P}_i(x) = H(x) \cdot P_i(x)$ accounts for the count in each bin.

5.4.2 Update Mean

The new mean μ_i' is the responsibility-weighted average of the data:

$$\mu_i' = \frac{\sum_{x \in X} x \cdot P_i(x)}{\sum_{x \in X} P_i(x)}$$

Interpretation: Points with higher responsibility $P_i(x)$ contribute more to the mean. This is like computing a weighted average where the weights reflect how strongly each point belongs to the component.

For histograms:

$$\mu_i' = \frac{\sum_{x \in X} x \cdot \hat{P}_i(x)}{\sum_{x \in X} \hat{P}_i(x)}$$

5.4.3 Update Variance

The new variance σ_i^2 (note: variance, not standard deviation) is:

$$\sigma_i'^2 = \frac{\sum_{x \in X} P_i(x) \cdot (x - \mu_i')^2}{\sum_{x \in X} P_i(x)}$$

Interpretation: This is the responsibility-weighted average squared deviation from the new mean. Points strongly associated with component i contribute more to its variance calculation.

For histograms:

$$\sigma_i'^2 = \frac{\sum_{x \in X} \hat{P}_i(x) \cdot (x - \mu_i')^2}{\sum_{x \in X} \hat{P}_i(x)}$$

After computing $\sigma_i'^2$, we take the square root to get the standard deviation: $\sigma_i' = \sqrt{\sigma_i'^2}$

5.5 EM Algorithm Summary

Input: Data X, number of components K (in our case, $K = 2$)

Initialization: Choose initial values for w_i, μ_i, σ_i

Repeat until convergence:

1. **E-Step:** Compute responsibilities $P_i(x)$ for all $x \in X$ and all components i
2. **M-Step:** Update parameters:

$$w_i' = \frac{1}{|X|} \sum_{x \in X} P_i(x) \tag{1}$$

$$\mu_i' = \frac{\sum_{x \in X} x \cdot P_i(x)}{\sum_{x \in X} P_i(x)} \tag{2}$$

$$\sigma_i'^2 = \frac{\sum_{x \in X} P_i(x) \cdot (x - \mu_i')^2}{\sum_{x \in X} P_i(x)} \tag{3}$$

3. **Check convergence:** Stop if parameters change by less than a threshold or maximum iterations reached

Output: Final parameters w_i, μ_i, σ_i

Important notes:

- EM guarantees the likelihood increases (or stays the same) with each iteration
- It converges to a local maximum, not necessarily the global maximum
- The algorithm provides the Maximum Likelihood Estimate (MLE) of the parameters given the observed data

6 Bringing It All Together: How GMM with EM Works

Let's walk through a complete example of how GMMs work in practice for image segmentation (foreground/background separation).

6.1 The Problem Setup

Task: Separate foreground and background pixels in a grayscale image

Assumption:

- Background pixels have lower intensities (darker), clustered around one mean
- Foreground pixels have higher intensities (brighter), clustered around another mean
- We don't know which pixels are foreground vs. background

Approach: Model the pixel intensity distribution as a mixture of two Gaussians

6.2 Step-by-Step Process

6.2.1 Step 0: Data Preparation

Collect all pixel intensities from the image into vector X. For a 100×100 image, $|X| = 10,000$.

For efficiency, we can compute the histogram $H(x)$ where x ranges from 0 to 255, giving us counts of how many pixels have each intensity.

6.2.2 Step 1: Initialization

Choose initial parameters. For example:

- Component 1 (background): $w_1 = 0.6$, $\mu_1 = 80$, $\sigma_1 = 20$
- Component 2 (foreground): $w_2 = 0.4$, $\mu_2 = 170$, $\sigma_2 = 20$

These initial guesses reflect our belief that more pixels are background (60%), backgrounds are darker (mean 80), and foregrounds are brighter (mean 170).

6.2.3 Step 2: E-Step (First Iteration)

For each pixel intensity value x, compute:

Component 1 likelihood:

$$L_1(x) = 0.6 \times \frac{1}{20\sqrt{2\pi}} e^{-\frac{(x-80)^2}{2(20)^2}}$$

Component 2 likelihood:

$$L_2(x) = 0.4 \times \frac{1}{20\sqrt{2\pi}} e^{-\frac{(x-170)^2}{2(20)^2}}$$

Total likelihood:

$$s_x = L_1(x) + L_2(x)$$

Responsibilities:

$$P_1(x) = \frac{L_1(x)}{s_x}, \quad P_2(x) = \frac{L_2(x)}{s_x}$$

Example calculation for a specific pixel with intensity $x = 90$:

$$L_1(90) = 0.6 \times \frac{1}{20\sqrt{2\pi}} e^{-\frac{(90-80)^2}{800}} \approx 0.6 \times 0.0193 \approx 0.0116$$

$$L_2(90) = 0.4 \times \frac{1}{20\sqrt{2\pi}} e^{-\frac{(90-170)^2}{800}} \approx 0.4 \times 0.0000014 \approx 0.00000056$$

$$s_{90} = 0.0116 + 0.00000056 \approx 0.0116$$

$$P_1(90) = \frac{0.0116}{0.0116} \approx 0.9999, \quad P_2(90) = \frac{0.00000056}{0.0116} \approx 0.0001$$

Interpretation: A pixel with intensity 90 is almost certainly (99.99%) from the background component, which makes sense since 90 is close to the background mean of 80.

6.2.4 Step 3: M-Step (First Iteration)

Using all the responsibilities computed in the E-step, update parameters:

Update weights:

$$w_1' = \frac{1}{10000} \sum_{x \in X} P_1(x)$$

If we sum all responsibilities and find $\sum P_1(x) = 6200$, then $w_1' = 0.62$.

Update means:

$$\mu_1' = \frac{\sum_{x \in X} x \cdot P_1(x)}{\sum_{x \in X} P_1(x)}$$

Suppose this weighted average yields $\mu_1' = 75$.

Update variances:

$$\sigma_1'^2 = \frac{\sum_{x \in X} P_1(x) \cdot (x - 75)^2}{\sum_{x \in X} P_1(x)}$$

Suppose this gives $\sigma_1'^2 = 450$, so $\sigma_1' = \sqrt{450} \approx 21.2$.

Similarly, we compute $w_2' = 0.38$, $\mu_2' = 175$, $\sigma_2' = 18$.

6.2.5 Step 4: Iterate

Replace the old parameters with the new ones and repeat the E-step and M-step. With each iteration:

- The parameters gradually adjust to better fit the data
- The log-likelihood of the data increases (EM's guarantee)
- Eventually, parameter changes become very small, indicating convergence

Typically, convergence occurs within 10-50 iterations for well-behaved data.

6.2.6 Step 5: Final Assignment and Thresholding

Once EM converges, we have final parameters. To make hard assignments (deciding if a pixel is foreground or background), we can:

Method 1: Maximum responsibility

- Assign pixel x to component i with highest $P_i(x)$
- If $P_1(x) > P_2(x)$, classify as background; otherwise, foreground

Method 2: Threshold at intersection point

Find the intensity value T where the two Gaussian densities (weighted by their mixing weights) are equal:

$$w_1 \cdot g(T|\mu_1, \sigma_1) = w_2 \cdot g(T|\mu_2, \sigma_2)$$

Taking logarithms and rearranging gives a quadratic equation:

$$\left(\frac{1}{\sigma_2^2} - \frac{1}{\sigma_1^2}\right) T^2 + 2\left(\frac{\mu_1}{\sigma_1^2} - \frac{\mu_2}{\sigma_2^2}\right) T + 2\ln\left(\frac{w_1}{w_2}\right) + 2\ln\left(\frac{\sigma_2}{\sigma_1}\right) - \frac{\mu_1^2}{\sigma_1^2} + \frac{\mu_2^2}{\sigma_2^2} = 0$$

Solving this quadratic equation gives the threshold(s). Pixels with intensity $x < T$ are classified as background, and $x \geq T$ as foreground.

Note: There can be two intersection points if the distributions overlap significantly. We choose the solution within the intensity range (typically 0-255 for images).

7 Solved Examples

Let's work through some concrete examples to solidify understanding.

7.1 Example 1: Simple 2-Component GMM with Small Dataset

Problem: We have data representing customer ages at two coffee shops (one near a university, one near a retirement community). The combined dataset is:

$$X = \{22, 23, 25, 27, 65, 67, 68, 70, 72\}$$

We suspect there are two groups (young and senior customers). Fit a 2-component GMM using one iteration of EM.

Solution:

7.1.1 Initialization

Looking at the data, we can guess:

- Component 1 (young): $w_1 = 0.5$, $\mu_1 = 25$, $\sigma_1 = 3$
- Component 2 (senior): $w_2 = 0.5$, $\mu_2 = 70$, $\sigma_2 = 3$

7.1.2 E-Step: Compute Responsibilities

For each data point x, compute $P_1(x)$ and $P_2(x)$.

For $x = 22$:

$$L_1(22) = 0.5 \times \frac{1}{3\sqrt{2\pi}} e^{-\frac{(22-25)^2}{2(3)^2}} = 0.5 \times \frac{1}{7.51} e^{-0.5} = 0.5 \times 0.0806 = 0.0403$$

$$L_2(22) = 0.5 \times \frac{1}{3\sqrt{2\pi}} e^{-\frac{(22-70)^2}{2(3)^2}} = 0.5 \times \frac{1}{7.51} e^{-128} \approx 0$$

$$s_{22} = 0.0403 + 0 \approx 0.0403$$

$$P_1(22) = \frac{0.0403}{0.0403} \approx 1.0, \quad P_2(22) = \frac{0}{0.0403} \approx 0$$

Following similar calculations for all data points:

x	$P_1(x)$	$P_2(x)$
22	1.0	0.0
23	1.0	0.0
25	1.0	0.0
27	1.0	0.0
65	0.0	1.0
67	0.0	1.0
68	0.0	1.0
70	0.0	1.0
72	0.0	1.0

7.1.3 M-Step: Update Parameters

Update weights:

$$w_1' = \frac{1}{9}(1+1+1+1+0+0+0+0+0) = \frac{4}{9} \approx 0.44$$

$$w_2' = \frac{1}{9}(0+0+0+0+1+1+1+1+1) = \frac{5}{9} \approx 0.56$$

Update means:

$$\mu_1' = \frac{22(1)+23(1)+25(1)+27(1)+65(0)+\ldots}{1+1+1+1+0+\ldots} = \frac{97}{4} = 24.25$$

$$\mu_2' = \frac{22(0)+\ldots+65(1)+67(1)+68(1)+70(1)+72(1)}{0+0+0+0+1+1+1+1+1} = \frac{342}{5} = 68.4$$

Update variances:

$$\sigma_1'^2 = \frac{(22-24.25)^2(1)+(23-24.25)^2(1)+(25-24.25)^2(1)+(27-24.25)^2(1)}{4} = \frac{14.74}{4} = 3.69$$

$$\sigma_1' = \sqrt{3.69} \approx 1.92$$

$$\sigma_2'^2 = \frac{(65-68.4)^2(1)+(67-68.4)^2(1)+(68-68.4)^2(1)+(70-68.4)^2(1)+(72-68.4)^2(1)}{5} = \frac{29.2}{5} = 5.84$$

$$\sigma_2' = \sqrt{5.84} \approx 2.42$$

7.1.4 Updated Parameters After First Iteration

- Component 1: $w_1 = 0.44$, $\mu_1 = 24.25$, $\sigma_1 = 1.92$
- Component 2: $w_2 = 0.56$, $\mu_2 = 68.4$, $\sigma_2 = 2.42$

Interpretation: The means have shifted slightly toward the data clusters, the weights now reflect that there are actually 5 senior customers vs. 4 young ones, and the standard deviations have decreased to better fit the tight clusters.

7.2 Example 2: GMM for Image Segmentation (Conceptual)

Problem: An image has a histogram with two clear peaks—one around intensity 50 and another around intensity 200. We want to use a GMM to automatically determine the threshold for separating foreground and background.

Given the histogram data (simplified):

Intensity	Count
45-55	3000
195-205	2000

After running EM, we obtain:

- Component 1 (background): $w_1 = 0.6$, $\mu_1 = 50$, $\sigma_1 = 15$
- Component 2 (foreground): $w_2 = 0.4$, $\mu_2 = 200$, $\sigma_2 = 10$

Find the threshold intensity that separates the two components.

Solution:

We need to find the intersection point where:

$$w_1 \cdot g(T|\mu_1, \sigma_1) = w_2 \cdot g(T|\mu_2, \sigma_2)$$

Substituting the Gaussian pdfs and taking natural logarithm:

$$\ln(w_1) + \ln\left(\frac{1}{\sigma_1\sqrt{2\pi}}\right) - \frac{(T-\mu_1)^2}{2\sigma_1^2} = \ln(w_2) + \ln\left(\frac{1}{\sigma_2\sqrt{2\pi}}\right) - \frac{(T-\mu_2)^2}{2\sigma_2^2}$$

Rearranging and substituting our values:

$$\ln(0.6) - \ln(15) - \frac{(T-50)^2}{2(15)^2} = \ln(0.4) - \ln(10) - \frac{(T-200)^2}{2(10)^2}$$

This simplifies to the quadratic equation:

$$\left(\frac{1}{100} - \frac{1}{225}\right)T^2 + 2\left(\frac{50}{225} - \frac{200}{100}\right)T + \ln(0.6) - \ln(0.4) - \ln(15) + \ln(10) - \frac{2500}{225} + \frac{40000}{100} = 0$$

Computing the coefficients:

- Coefficient of T^2: $\frac{1}{100} - \frac{1}{225} = \frac{225-100}{22500} = \frac{125}{22500} = \frac{1}{180}$
- Coefficient of T: $2\left(\frac{50}{225} - \frac{200}{100}\right) = 2\left(\frac{2}{9} - 2\right) = 2\left(-\frac{16}{9}\right) = -\frac{32}{9}$
- Constant term: $\ln\left(\frac{0.6}{0.4}\right) + \ln\left(\frac{10}{15}\right) - \frac{2500}{225} + 400 = \ln(1.5) + \ln(2/3) - 11.11 + 400 \approx 389.30$

The quadratic equation becomes:

$$\frac{1}{180}T^2 - \frac{32}{9}T + 389.30 = 0$$

Multiplying by 180:

$$T^2 - 640T + 70074 = 0$$

Using the quadratic formula:

$$T = \frac{640 \pm \sqrt{640^2 - 4(70074)}}{2} = \frac{640 \pm \sqrt{409600 - 280296}}{2} = \frac{640 \pm \sqrt{129304}}{2} = \frac{640 \pm 359.6}{2}$$

This gives two solutions: $T_1 = 499.8$ and $T_2 = 140.2$.

Since our components have means at 50 and 200, the threshold $T = 140.2$ makes sense—it's between the two peaks.

Threshold: $T \approx 140$

Interpretation: Pixels with intensity less than 140 should be classified as background (component 1), and pixels with intensity 140 or greater should be classified as foreground (component 2).

8 Key Takeaways

8.1 When to Use GMMs

Use GMMs when:

- Data appears to come from multiple overlapping populations

- You need probabilistic (soft) cluster assignments
- The number of clusters is known or can be estimated
- Data is approximately Gaussian within each cluster

Avoid GMMs when:

- Clusters have very non-Gaussian shapes (consider kernel density estimation)
- The number of clusters is completely unknown (try methods with automatic cluster selection)
- Data is high-dimensional with many irrelevant features (curse of dimensionality)

8.2 Important Points to Remember

1. **Initialization matters:** EM converges to local maxima, so try multiple random initializations or use k-means for initialization
2. **Convergence criteria:** Typically stop when:
   - Parameter changes are below a threshold (e.g., $|\mu_i' - \mu_i| < 0.01$)
   - Log-likelihood improvement is negligible
   - Maximum iterations reached
3. **Model selection:** For unknown number of components, use:
   - Bayesian Information Criterion (BIC)
   - Akaike Information Criterion (AIC)
   - Cross-validation
   - Elbow method on log-likelihood
4. **Computational efficiency:**
   - For images, use histograms instead of individual pixels
   - Vectorize operations when implementing
   - Consider diagonal covariance matrices for high dimensions
5. **Theoretical guarantees:**
   - EM monotonically increases likelihood (never decreases)
   - Converges to a local maximum (not necessarily global)
   - Provides maximum likelihood estimates

8.3 Common Applications

- **Image segmentation:** Separating foreground/background or multiple regions
- **Anomaly detection:** Points with low probability under the mixture are outliers
- **Density estimation:** Modeling complex probability distributions
- **Feature learning:** Learning representations in deep learning
- **Speaker recognition:** Modeling acoustic features with multiple Gaussian components
- **Financial modeling:** Modeling returns with different market regimes

8.4 Extensions and Variations

- **Full covariance GMMs:** Allow correlations between features (not just diagonal covariance)
- **Bayesian GMMs:** Put priors on parameters for regularization
- **Infinite GMMs:** Use Dirichlet processes to automatically determine number of components
- **Multivariate GMMs:** Extend to multiple features (vector-valued observations)